BRADEN GALLAGHER

College Station, TX 77840 | (469) 667-7917 | bradengallagher@tamu.edu | linkedin.com/in/braden-gallagher |
bmoney2029.github.io/portfolio

EDUCATION

Texas A&M University College Station, TX
B.S. in Mechanical Engineering - GPA: 3.84 *Expected May 2029*

ENGINEERING EXPERIENCE

Texas A&M Rocket Engine Design (RED) College Station, TX
Chief Engineer - Elysium II Liquid Rocket Engine *August 2026 – Present*

- Lead technical authority for the Elysium II engine program: direct every engine subteam and own cross subteam integration so all sub-assemblies meet at their interfaces and function as one system.
- Provide final engineering checkoff on hardware, procedures, and system checkouts, reviewing test readiness and signing off before any test is executed.
- Run cross subteam design reviews and resolve interface conflicts to hold the program to design intent and schedule.

Thrust Chamber Assembly Lead *January 2026 – August 2026*

- Directed a multidisciplinary team of 9 engineers, overseeing technical workflows and part timelines.
- Redesigned the Elysium II fuel injector and manifold for in house manufacturability, then manufactured and assembled the completed injector assembly, bringing the design from CAD to finished, test ready hardware.
- Conducted thermal analysis to implement cooling for a chamber at ~3000 K and 1,500 lbf thrust.
- Manufactured torch igniter for the Ragnarok engine to tolerances within 0.005 in.

Albireo LLC Fort Worth, TX
Engineering Intern *May 2026 – Present*

- Project lead on a robotic control system (electronics, stepper drives) for an astronomical observation platform.
- Created control circuit schematics in KiCad and fabricated custom PCBs in house using CNC milling.
- Built a remote CNC with a custom Arduino based control interface, expanding in house PCB capability.

PROJECTS

RC Aircraft — Design, Test & Build Personal Project
Propulsion Testing, Aerodynamic Simulation & Additive Manufacturing *June 2026 – Present*

- Designed and built a load-cell equipped thrust stand to measure real thrust across propeller and motor combinations.
- Running XFLR5 aerodynamic analysis for wing sizing and stability, and ANSYS Fluent CFD correlated against measured thrust data.
- Translating validated aero and propulsion data into a full SolidWorks airframe, 3D printed for flight testing.

Turtle Robotics Team College Station, TX
Hatchling Competition *January 2026 – May 2026*

- Developed Python and C++ navigation algorithms for a microcontroller-based robot, placing 3rd of 34 teams organization-wide.
- Engineered mechanical components in SolidWorks and 3D printing to optimize robot durability.

TECHNICAL SKILLS

Hardware & Machining: FEDC Red and Yellow Badge Certified, Manual Mill & Lathe, CNC Lathe (in training), Welding, 3D Printing, PCB Fabrication (CNC-milled)

Software: SolidWorks, KiCad, ANSYS Fluent (CFD), XFLR5, FEA, Python, GitHub

EXTRACURRICULARS & LEADERSHIP

Aggie Club of Engineers: Service Team (Sept. 2025 – Present): Raised $5,000+ for Camp Hope, a PTSD Foundation of America veterans' program; 15+ volunteer service hours.

John Paul II High School Tennis: Varsity Team Captain (2022 – 2025): 2x State Finalist; led team to back-to-back state championship appearances in doubles.